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Mini Project — Phase II

Implementing an Online Shopping Cart System

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1 Abstract

This project presents the design and implementation of a web-based Shopping Cart Management System intended to demonstrate practical applications of Database Management System (DBMS) concepts. The system enables users to browse items, add or remove products from a virtual shopping cart, update quantities, and perform a consolidated checkout operation. The primary focus is on database modelling, integrity enforcement, and transaction management rather than building a full-scale commercial e-commerce platform.

The design extends the bookstore Entity–Relationship (E-R) schema of Exercise 6.21 (Silberschatz, *Database System Concepts*, 7th ed.) to incorporate additional entities such as *Cart*, *Order*, *Inventory*, and *Payment*, ensuring proper relationships, normalisation, and constraint enforcement. The relational schema is implemented using Oracle 21c XE with primary keys, foreign keys, check constraints, sequences, triggers, and stored procedures. A Java JDBC layer demonstrates database connectivity and calls to PL/SQL procedures for cart operations, stock verification, and atomic order placement.

2 Introduction

The rapid growth of e-commerce has made efficient shopping cart systems a cornerstone of modern retail infrastructure. Managing product catalogues, customer sessions, inventory, and atomic transactions demands a well-designed relational database backend capable of enforcing integrity at every layer.

This project implements a Shopping Cart Management System for an online bookstore, built atop Oracle 21c XE using JDBC and JavaFX. The system extends the bookstore Entity–Relationship schema of Exercise 6.21 (Silberschatz, *Database System Concepts*, 7th ed.) to support a complete purchase lifecycle: browsing, cart management, stock verification, and atomic checkout.

The primary objective is not to replicate a commercial platform but to demonstrate rigorous application of DBMS principles — normalisation, constraint enforcement, stored procedures, triggers, and transaction management — within a working, interactive system.

3 Problem Statement

3.1 Background

Design and implement a shopping cart system that lets shoppers collect items into a shopping cart and purchase them together. The system is built by extending the bookstore E-R schema of **Exercise 6.21, Chapter 6** (Korth, 6th Edition). Stock availability must be verified at checkout, and out-of-stock situations must be handled gracefully.

3.2 Data Requirements

The system must maintain the following data:

- **Books**: identified by ISBN; storing title, publication year, price, genre, author, and publisher.
- **Authors**: name, address, URL.
- **Publishers**: name, address, phone, URL.
- **Customers**: unique email, name, address, phone, hashed password, registration timestamp.
- **Warehouses**: code, address, phone.
- **Inventory**: per-warehouse stock count for each book.
- **Cart**: links a customer to a set of items; tracks status (ACTIVE, CHECKED_OUT, ABANDONED).
- **Cart Items**: book + quantity within a cart.
- **Orders**: placed from a cart; captures total amount and order status.
- **Order Items**: snapshot of each book's price and quantity at order time.
- **Payments**: method, amount, and status for each order.

3.3 Functional Requirements

1. Browse all available books with author/publisher details.
2. Add, update quantity, and remove books from an active cart.
3. Check total stock availability before and during checkout.
4. Atomically place an order from a cart using a transaction; rollback on failure.
5. Automatically decrement inventory on order placement (trigger).
6. Mark a cart as CHECKED_OUT when an order is placed (trigger).
7. Prevent negative inventory at all times (trigger + constraint).
8. Record payment alongside each order.
9. Support complex reporting: revenue by publisher, demand forecasting, order receipts, and customer analytics.

4 Methodology

4.1 Development Approach

The project followed an incremental design process: the database schema was designed and normalised first, followed by DDL implementation, then PL/SQL logic, and finally the Java JDBC frontend. Each layer was tested independently before integration.

4.2 Tools and Technologies

- **Database:** Oracle 21c XE (XEPDB1) for schema hosting, constraint enforcement, and PL/SQL execution.
- **Connectivity:** Oracle JDBC Type-4 thin driver (`ojdbc11.jar`) for Java–database communication.
- **Frontend:** JavaFX for the graphical user interface, providing a dark-themed tabbed dashboard.
- **Build Tool:** Apache Maven for dependency management and compilation.
- **IDE:** IntelliJ IDEA for development and debugging.

4.3 Database Design Process

The original Exercise 6.21 E-R schema was analysed to identify entities relevant to a shopping cart context. Extensions were then designed to cover the full purchase lifecycle, including Cart, Inventory, Orders, Order Items, and Payment entities. Each table was verified against 3NF before DDL was written.

4.4 Implementation Sequence

1. Schema design and normalisation.
2. DDL: table definitions, constraints, and sequences.
3. Sample data insertion across all 11 tables.
4. PL/SQL: functions, procedures, and triggers.
5. JDBC layer: connection setup, query execution, and procedure calls.
6. JavaFX UI: one tab per core operation, with a live console log.
7. Testing against all scenarios in Table 5.

4.5 Transaction Strategy

All multi-step write operations — particularly checkout — use manual transaction control (`setAutoCommit(false)`) with explicit `COMMIT` on success and `ROLLBACK` in the exception handler, ensuring atomicity across order placement, inventory decrement, and payment recording.

5 E-R Diagram and Schema Extension

5.1 Original E-R Schema (Exercise 6.21)

The original bookstore schema from Silberschatz et al. contains the following entities and relationships:

- **Entities**: *author* (name, address, URL), *book* (ISBN, title, year, price), *publisher* (name, address, phone, URL), *customer* (email, name, address, phone), *shopping_basket* (basket_id), *warehouse* (code, address, phone).
- **Relationships**: *written_by* (book–author), *published_by* (book–publisher), *basket_of* (basket–customer), *contains* (basket–book, with attribute *number*), *stocks* (warehouse–book, with attribute *number*).

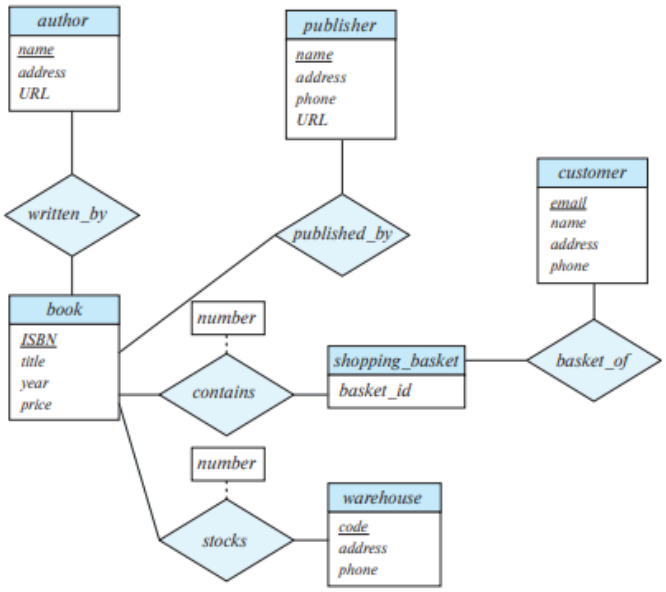

Figure 1: Original E-R schema from Exercise 6.21 (Silberschatz et al.)

5.2 Extensions Made

Table 1 summarises the extensions applied to the original schema.

Table 1: Schema extensions beyond Exercise 6.21

Original	Change	Reason
book	Added *genre*, *description*	Richer browsing
customer	Added *password_hash*, *created_at*	Auth + analytics
shopping_basket	Renamed to *cart*; added *status*, *created_at*	Lifecycle tracking
contains	Became *CART_ITEMS* table	Proper relation
stocks	Became *INVENTORY* table	Warehouse tracking
—	New: *ORDERS*	Order management
—	New: *ORDER_ITEMS*	Per-order snapshot
—	New: *PAYMENT*	Payment tracking

6 Relational Schema and Normalisation

6.1 Relational Tables

AUTHOR(*name*, *address*, *url*)
PUBLISHER(*name*, *address*, *phone*, *url*)
BOOK(*isbn*, *title*, *year*, *price*, *genre*, *description*, *author_name*, *publisher_name*)
CUSTOMER(*cust_id*, *email*, *name*, *address*, *phone*, *password_hash*, *created_at*)
WAREHOUSE(*code*, *address*, *phone*)
INVENTORY(*warehouse_code*, *isbn*, *quantity*)
CART(*cart_id*, *cust_id*, *status*, *created_at*, *updated_at*)
CART_ITEMS(*cart_id*, *isbn*, *quantity*)
ORDERS(*order_id*, *cust_id*, *cart_id*, *order_date*, *total_amount*, *status*)
ORDER_ITEMS(*order_id*, *isbn*, *quantity*, *price_at_order*)
PAYMENT(*payment_id*, *order_id*, *amount*, *method*, *payment_date*, *status*)

(Underlined attributes are primary keys; foreign key references follow from context.)

6.2 Normalisation

All tables satisfy **Third Normal Form (3NF) / BCNF**:

- **1NF**: All attributes are atomic. No multi-valued columns.
- **2NF**: No partial dependencies exist. In `ORDER_ITEMS` the non-key attribute `price_at_order` depends on the full composite key {order_id, isbn}, not a subset.
- **3NF**: No transitive dependencies. `BOOK` stores `author_name` and `publisher_name` as FKs only — author details live entirely in `AUTHOR`.

A deliberate design choice: `ORDER_ITEMS.price_at_order` snapshots the book price at time of purchase. Although `BOOK.price` exists, this is not a transitive dependency because the historical price is a distinct fact from the current price.

7 DDL — Table Definitions and Constraints

The following listings show representative DDL. Full code is in `schema_setup.sql`.

Listing 1: BOOK and INVENTORY table definitions

```
CREATE TABLE BOOK (
    isbn            VARCHAR2(20)  NOT NULL,
    title           VARCHAR2(255) NOT NULL,
    year            NUMBER(4)     NOT NULL,
    price           NUMBER(8,2)   NOT NULL,
    genre           VARCHAR2(50),
    author_name     VARCHAR2(100) NOT NULL,
    publisher_name  VARCHAR2(100) NOT NULL,
```

```
    CONSTRAINT pk_book        PRIMARY KEY (isbn),
    CONSTRAINT fk_book_author FOREIGN KEY (author_name)
                              REFERENCES AUTHOR(name),
    CONSTRAINT fk_book_pub    FOREIGN KEY (publisher_name)
                              REFERENCES PUBLISHER(name),
    CONSTRAINT chk_book_price CHECK (price > 0),
    CONSTRAINT chk_book_year  CHECK (year BETWEEN 1000 AND 2100)
);

CREATE TABLE INVENTORY (
    warehouse_code VARCHAR2(20) NOT NULL,
    isbn           VARCHAR2(20) NOT NULL,
    quantity       NUMBER(6)    DEFAULT 0 NOT NULL,
    CONSTRAINT pk_inventory     PRIMARY KEY (warehouse_code, isbn),
    CONSTRAINT fk_inv_warehouse FOREIGN KEY (warehouse_code)
                                REFERENCES WAREHOUSE(code),
    CONSTRAINT fk_inv_book      FOREIGN KEY (isbn)
                                REFERENCES BOOK(isbn),
    CONSTRAINT chk_inv_qty      CHECK (quantity >= 0)
);
```

Listing 2: CART and ORDERS with status constraints

```
CREATE TABLE CART (
    cart_id    NUMBER       DEFAULT SEQ_CART.NEXTVAL NOT NULL,
    cust_id    NUMBER       NOT NULL,
    status     VARCHAR2(20) DEFAULT 'ACTIVE' NOT NULL,
    created_at DATE         DEFAULT SYSDATE,
    CONSTRAINT pk_cart       PRIMARY KEY (cart_id),
    CONSTRAINT fk_cart_cust  FOREIGN KEY (cust_id)
                             REFERENCES CUSTOMER(cust_id),
    CONSTRAINT chk_cart_status
        CHECK (status IN ('ACTIVE','CHECKED_OUT','ABANDONED'))
);
```

8 SQL Queries

8.1 Basic Queries

Listing 3: Q1: All books with author and publisher

```
SELECT b.isbn, b.title, b.year, b.price,
       a.name AS author, p.name AS publisher
FROM   BOOK b
JOIN   AUTHOR    a ON a.name = b.author_name
JOIN   PUBLISHER p ON p.name = b.publisher_name
ORDER  BY b.title;
```

Listing 4: Q2: Active cart contents for a customer

```
SELECT b.title, b.price, ci.quantity,
```

```
        (b.price * ci.quantity) AS line_total
FROM    CART c
JOIN    CART_ITEMS ci ON ci.cart_id = c.cart_id
JOIN    BOOK b        ON b.isbn     = ci.isbn
WHERE   c.cust_id = 1 AND c.status = 'ACTIVE';
```

Listing 5: Q3: Total stock per book across all warehouses

```
SELECT b.title, SUM(i.quantity) AS total_stock
FROM   INVENTORY i
JOIN   BOOK b ON b.isbn = i.isbn
GROUP  BY b.title
ORDER  BY total_stock DESC;
```

8.2 Complex Queries

Listing 6: Q9: Stock availability check for cart items

```
SELECT  b.title,
        ci.quantity              AS qty_wanted,
        NVL(SUM(i.quantity),0)   AS total_available,
        CASE
            WHEN NVL(SUM(i.quantity),0) >= ci.quantity THEN 'IN
               STOCK'
            WHEN NVL(SUM(i.quantity),0) > 0            THEN
               'PARTIAL'
            ELSE                                            'OUT OF
               STOCK'
        END AS availability
FROM    CART_ITEMS ci
JOIN    BOOK       b  ON b.isbn = ci.isbn
LEFT JOIN INVENTORY i ON i.isbn = ci.isbn
WHERE   ci.cart_id = 1
GROUP   BY b.title, ci.quantity;
```

Listing 7: Q8: Active cart summary with totals

```
SELECT  cu.name AS customer, c.cart_id,
        COUNT(ci.isbn)             AS distinct_titles,
        SUM(ci.quantity)           AS total_books,
        SUM(b.price * ci.quantity) AS cart_value
FROM    CART c
JOIN    CUSTOMER   cu ON cu.cust_id = c.cust_id
JOIN    CART_ITEMS ci ON ci.cart_id = c.cart_id
JOIN    BOOK       b  ON b.isbn     = ci.isbn
WHERE   c.status = 'ACTIVE'
GROUP   BY cu.name, c.cart_id
ORDER   BY cart_value DESC;
```

Listing 8: Q11: Revenue per publisher (complex aggregation)

```
SELECT  p.name AS publisher,
```

```
        COUNT(DISTINCT o.order_id)           AS total_orders,
        SUM(oi.quantity)                     AS books_sold,
        SUM(oi.price_at_order * oi.quantity) AS revenue
FROM    ORDER_ITEMS oi
JOIN    BOOK        b  ON b.isbn     = oi.isbn
JOIN    PUBLISHER   p  ON p.name     = b.publisher_name
JOIN    ORDERS      o  ON o.order_id = oi.order_id
WHERE   o.status IN ('PLACED','SHIPPED','DELIVERED')
GROUP   BY p.name
ORDER   BY revenue DESC;
```

9 PL/SQL Procedures, Functions, and Triggers

9.1 Functions

Listing 9: get_total_stock: total stock across all warehouses

```
CREATE OR REPLACE FUNCTION get_total_stock(p_isbn IN VARCHAR2)
RETURN NUMBER IS
    v_total NUMBER := 0;
BEGIN
    SELECT NVL(SUM(quantity), 0) INTO v_total
    FROM   INVENTORY WHERE isbn = p_isbn;
    RETURN v_total;
END get_total_stock;
```

9.2 Procedures

Listing 10: add_to_cart: add a book to a customer's active cart

```
CREATE OR REPLACE PROCEDURE add_to_cart(
    p_cust_id  IN NUMBER,
    p_isbn     IN VARCHAR2,
    p_quantity IN NUMBER DEFAULT 1
) AS
    v_cart_id NUMBER;
    v_stock   NUMBER;
BEGIN
    v_stock := get_total_stock(p_isbn);
    IF v_stock < p_quantity THEN
        RAISE_APPLICATION_ERROR(-20010,
            'Insufficient stock. Available: ' || v_stock);
    END IF;
    -- get or create active cart, then upsert cart item
    ...
    COMMIT;
END add_to_cart;
```

Listing 11: checkout: atomic order placement with rollback

```oracle
CREATE OR REPLACE PROCEDURE checkout(
    p_cust_id        IN  NUMBER,
    p_payment_method IN  VARCHAR2,
    p_order_id       OUT NUMBER
) AS BEGIN
    -- 1. Validate cart not empty
    -- 2. Check all items in stock
    -- 3. INSERT into ORDERS
    -- 4. INSERT into ORDER_ITEMS  (triggers fire here)
    -- 5. INSERT into PAYMENT
    COMMIT;
EXCEPTION
    WHEN OTHERS THEN ROLLBACK; RAISE;
END checkout;
```

9.3 Triggers

Table 2: Summary of triggers

Trigger	Event	Action
trg_decrement_stock	AFTER INSERT on ORDER_ITEMS	Decrement INVENTORY
trg_checkout_cart	AFTER INSERT on ORDERS	Mark cart CHECKED_OUT
trg_cart_updated	AFTER DML on CART_ITEMS	Update cart timestamp
trg_prevent_neg_stock	BEFORE UPDATE on INVENTORY	Reject negative qty
trg_lock_checked_out	BEFORE DML on CART_ITEMS	Reject non-ACTIVE cart

Listing 12: trg_decrement_stock trigger

```oracle
CREATE OR REPLACE TRIGGER trg_decrement_stock
AFTER INSERT ON ORDER_ITEMS
FOR EACH ROW
BEGIN
    UPDATE INVENTORY
    SET    quantity = quantity - :NEW.quantity
    WHERE  isbn = :NEW.isbn;
END;
```

10 Java DB Connectivity

The file `ShoppingCartDB.java` demonstrates Oracle JDBC connectivity using the Type-4 thin driver (`ojdbc11.jar`).

Listing 13: JDBC connection setup

```java
private static final String URL =
    "jdbc:oracle:thin:@localhost:1521/XEPDB1";

public static Connection getConnection() throws SQLException {
```

```
    return DriverManager.getConnection(URL, USER, PASSWORD);
}
```

Listing 14: Calling the checkout PL/SQL procedure

```
String sql = "BEGIN checkout(?, ?, ?); END;";
CallableStatement cs = con.prepareCall(sql);
cs.setInt(1, custId);
cs.setString(2, paymentMethod);
cs.registerOutParameter(3, Types.NUMERIC); // OUT p_order_id
cs.execute();
int orderId = cs.getInt(3);
```

Key JDBC patterns demonstrated:

- `Statement` for simple SELECT queries.
- `PreparedStatement` for parameterised queries (prevents SQL injection).
- `CallableStatement` for PL/SQL procedure and function calls.
- Manual `con.setAutoCommit(false)` with explicit `COMMIT`/`ROLLBACK`.

11 Transaction Management

The checkout operation is the most critical transaction in the system. It follows an all-or-nothing strategy:

1. Validate that the cart contains at least one item.
2. For every cart item, verify that total warehouse stock $\geq$ quantity requested. If any item fails, raise an error before any `INSERT`.
3. Insert one row into `ORDERS`.
4. Insert rows into `ORDER_ITEMS`; triggers fire automatically to decrement `INVENTORY` and mark the cart `CHECKED_OUT`.
5. Insert one row into `PAYMENT`.
6. `COMMIT` — all five steps succeed atomically.
7. `ROLLBACK` in the `WHEN OTHERS` exception handler if any step fails.

This guarantees that partial orders (e.g., payment recorded but inventory not decremented) can never occur.

12 Sample Data

Table 3: Sample books in the system

ISBN	Title	Year	Price (Rs.)
978-0-07-352332-3	Database System Concepts	2019	899
978-0-13-468599-1	Operating System Concepts	2018	799
978-0-13-235088-4	Clean Code	2008	649
978-0-13-212227-1	Computer Networks	2021	749
978-0-06-231609-7	Sapiens	2015	499
978-0-06-244558-9	Homo Deus	2017	549

Table 4: Warehouses and total inventory

Code	Location	Books Stocked
WH-MUM	Andheri East, Mumbai	3 titles
WH-BLR	Whitefield, Bengaluru	3 titles
WH-DEL	Okhla Phase II, New Delhi	3 titles

13 UI Design

The application is built using **JavaFX** with a dark-themed dashboard interface. It connects to Oracle 21c XE via JDBC and exposes all six core operations as tabs. A persistent console panel at the bottom logs every DB operation in real time, and a status indicator in the header confirms the active Oracle connection (`XEPDB1`).

13.1 Books Listing

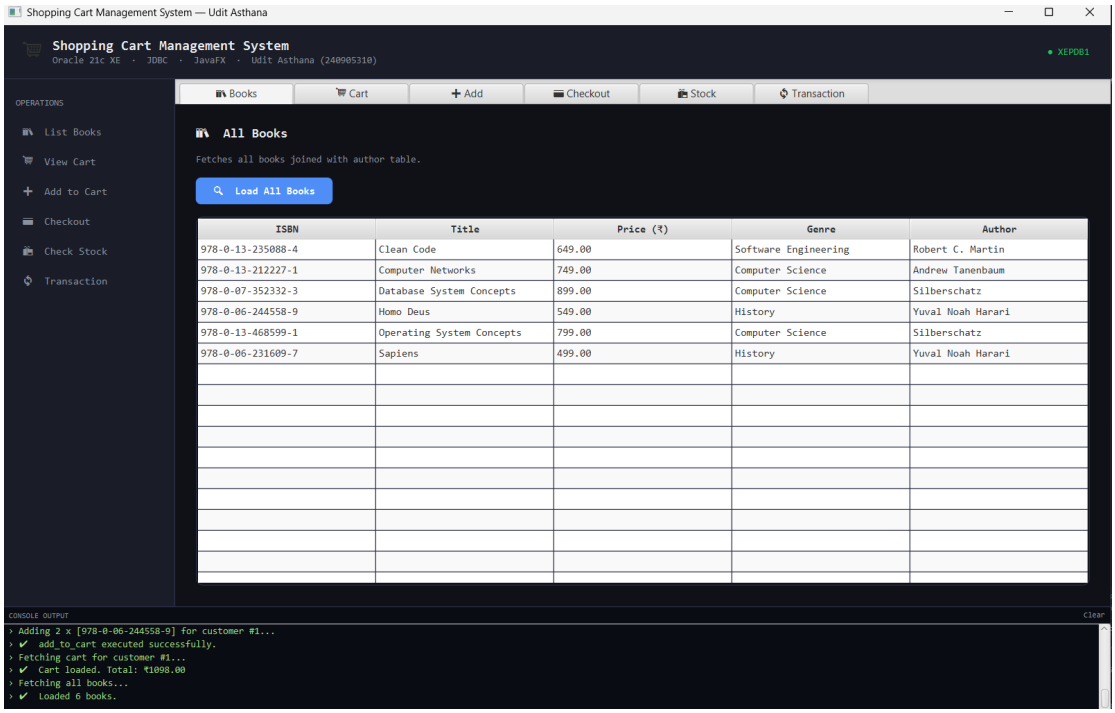

Figure 2: Books tab — all 6 books fetched via `SELECT ... JOIN AUTHOR`. Console shows `Loaded 6 books.`

13.2 View Cart

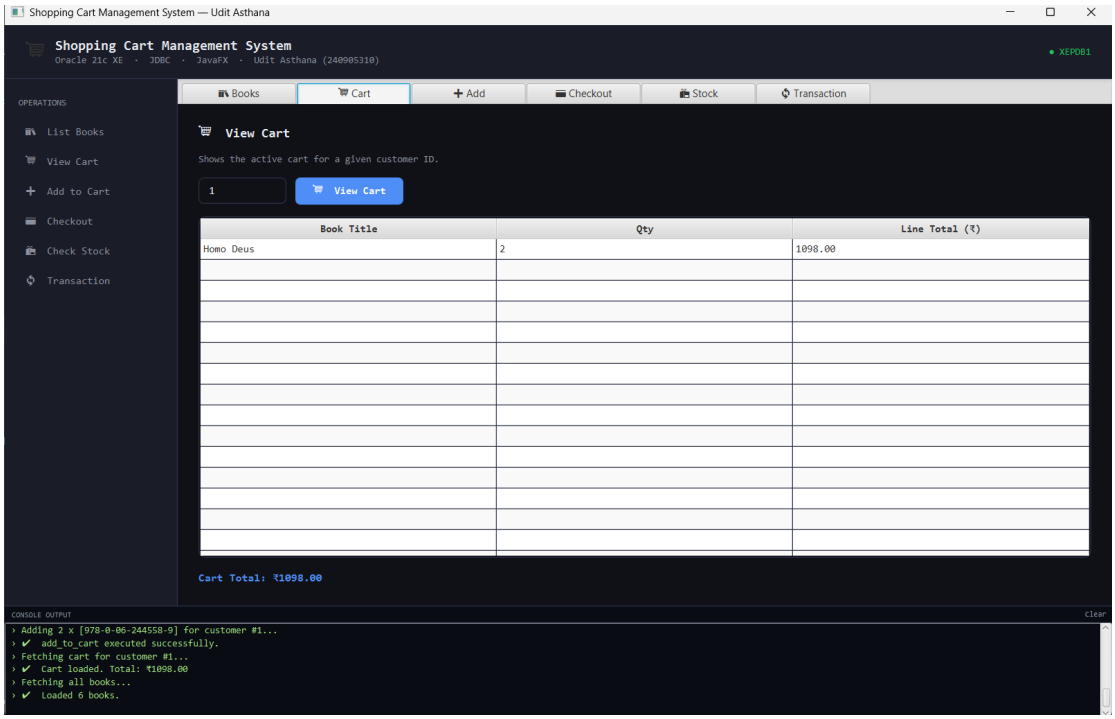

Figure 3: Cart tab — active cart for Customer #1 showing *Homo Deus* (qty 2, line total 1098.00). Uses `PreparedStatement` with customer ID as parameter.

13.3 Add to Cart

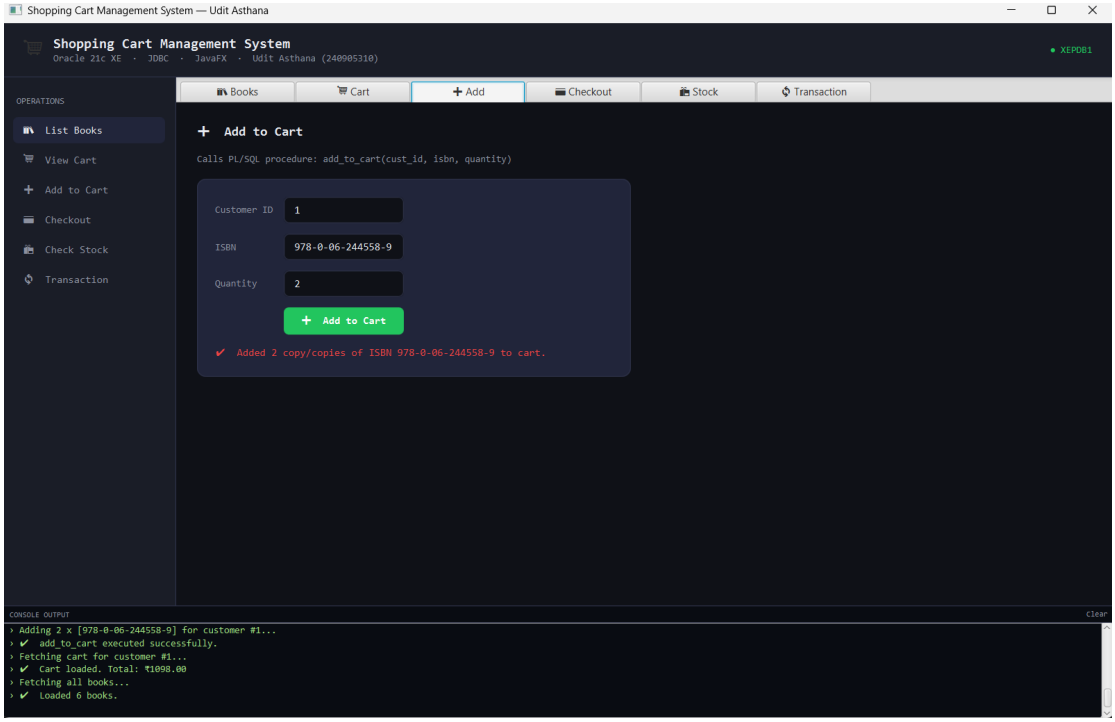

Figure 4: Add to Cart tab — Customer #1 adds 2 copies of ISBN `978-0-06-244558-9`. Calls PL/SQL procedure `add_to_cart(cust_id, isbn, quantity)` via `CallableStatement`.

13.4 Checkout

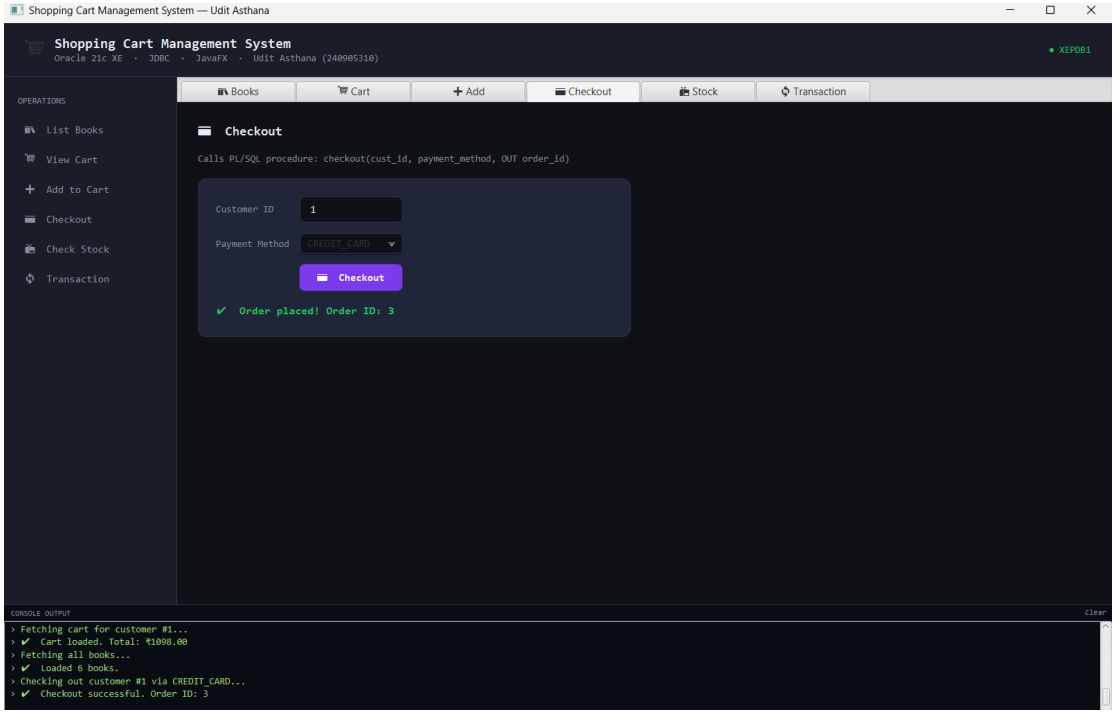

Figure 5: Checkout tab — Customer #1 checks out via `CREDIT_CARD`. Order ID 3 returned from the `checkout` PL/SQL procedure OUT parameter. Console confirms `Checkout successful.`

13.5 Check Stock

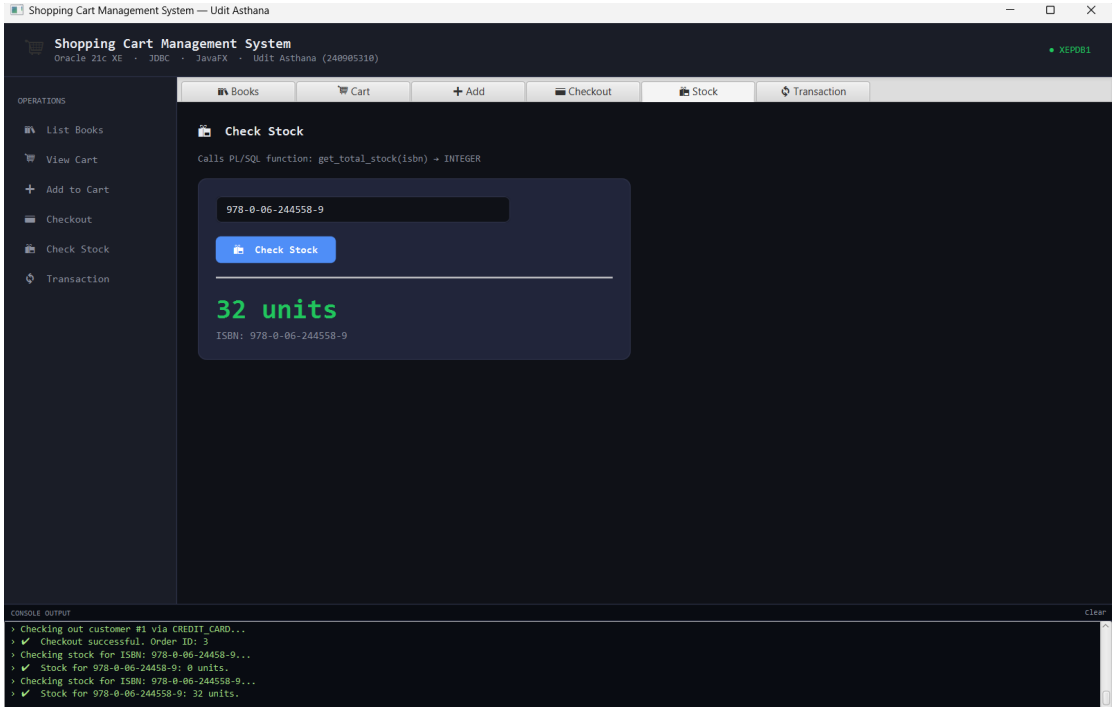

Figure 6: Stock tab — calls PL/SQL function `get_total_stock(isbn)` and displays result. ISBN `978-0-06-244558-9` shows 32 units remaining after the checkout above decremented inventory via trigger.

13.6 Transaction Demo

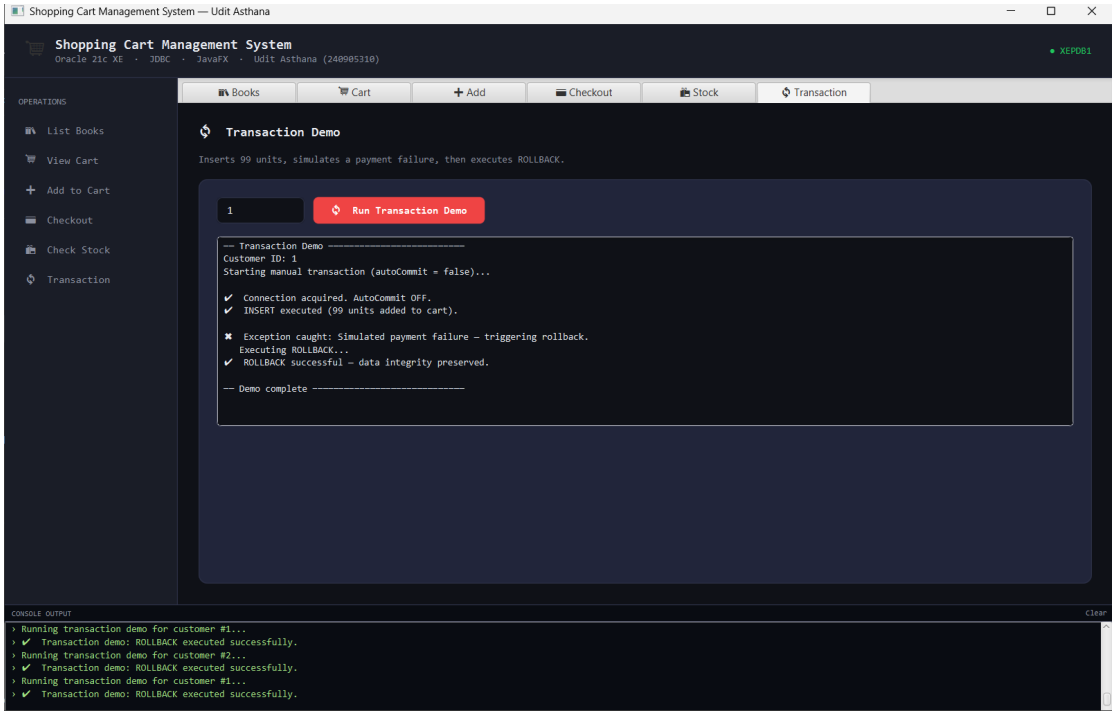

Figure 7: Transaction Demo tab — inserts 99 units into cart, simulates a payment failure, then executes `ROLLBACK`. Demonstrates `con.setAutoCommit(false)` with explicit rollback on exception. Data integrity is preserved.

14 Testing Scenarios

Table 5: Test cases and expected outcomes

Test Case	Input	Expected Result
Add to cart (valid)	ISBN + qty in stock	Item added, cart updated
Add to cart (out of stock)	qty > total stock	ORA-20010 raised
Checkout (valid)	Active cart, items available	Order placed, stock decremented
Checkout (empty cart)	Cart with 0 items	ORA-20030 raised
Checkout (partial stock)	One item insufficient	ORA-20031 raised, rollback
Modify checked-out cart	DML on cart_id=2	ORA-20060 raised (trigger)
Negative stock update	SET quantity = -1	ORA-20050 raised (trigger)
Duplicate email registration	Same email twice	ORA-00001 (unique constraint)

15 Conclusion

This project successfully demonstrates the design and implementation of a Shopping Cart Management System grounded in rigorous DBMS principles. Starting from the Exercise 6.21 bookstore E-R schema, the system was extended to support a complete purchase lifecycle — from cart creation through checkout to payment recording.

Key outcomes achieved:

- A fully normalised (3NF) relational schema with 11 tables.
- 15 SQL queries spanning basic retrieval to complex multi-table aggregation.
- 4 PL/SQL stored procedures and 2 functions encapsulating core business logic.
- 5 triggers enforcing referential and business-rule integrity automatically.
- An atomic, rollback-safe checkout transaction preventing partial order anomalies.
- A Java JDBC layer demonstrating `PreparedStatement`, `CallableStatement`, and manual transaction control.

15.1 Limitations

- The system supports a single active cart per customer; concurrent sessions are not handled.
- Authentication is not implemented in the UI; customer ID is entered manually.
- The JavaFX frontend is desktop-only and not suitable for web or mobile deployment.
- Inventory is decremented by trigger at order insertion but is not reserved during cart session, allowing race conditions under concurrent load.

15.2 Future Work

- Implement session-based authentication with hashed password verification.
- Add optimistic locking or inventory reservation during cart checkout to handle concurrent users.
- Migrate the frontend to a REST API backend with a web interface.
- Extend reporting queries into a live analytics dashboard within the UI.

References

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